

IE3101 — Enterprise Network and Information Systems Security
Year 3
Semester 01
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Year 3 Semester 1
2026

Assignment Title	Network Security Assessment and Secure Network Design
Learning outcomes covered	LO 1: Classify Network Infrastructure Security Access Control and their relevance across network layers LO 2: Describing risks to computer systems and various attack strategies LO 3: Analyze network security monitoring practices and incident response procedures
Assignment Mode	Group Assignment
Maximum Marks	100
Contribution to the Final Grade	20%
Date published	16/09/2026
Deadline for submissions	02/10/2026
Mode of Submission	Report and presentation slide

Assignment description
1. Assignment Context

Each group must use the same real client selected for the Information Systems Project Management module. For IE3101, the group will act as a Network and Information Security Consulting Team and assess the client's current or anticipated network environment. The main objective is to identify security weaknesses and risks, conduct a structured vulnerability and risk assessment, align major treatments with ISO/IEC 27001:2022, and propose a secure and scalable network architecture.

2. Select ONE Client Scenario
i. Option A – Existing Network Infrastructure

Use this option when the client already operates a network that can be studied. Document the current As-Is environment, identify vulnerabilities, security gaps and risks, and propose an improved To-Be secure network.

ii. Option B – New or Expanded Network

Use this option when the client has a limited network or plans to expand operations, branches, users, devices, servers, cloud services, IoT or wireless services. Identify current limitations and future requirements, assess foreseeable vulnerabilities and risks, and design a secure and scalable network before implementation.

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3. Required Assignment Tasks
Task 1 – Client and Network Environment Analysis

Briefly introduce the client and the business operations that depend on network and information systems. State the selected option and justify it. Identify relevant users, departments, devices, servers/services, Internet connectivity, wireless and remote-access requirements, cloud services, critical applications, sensitive information and expected future growth.

Task 2 – Current/Initial Network Architecture

Option A: develop an As-Is Network Diagram showing relevant Internet connectivity, perimeter security, routers, switches, VLANs/segments, servers, wireless infrastructure, endpoints and cloud/external services, together with existing security controls. Option B: develop a baseline/current-state diagram and clearly identify the future network and scalability requirements. Designs must reflect the client's real business needs.

Task 3 – Initial Security and Vulnerability Identification

Identify potential weaknesses in the existing or proposed environment. Consider network segmentation, Internet exposure, perimeter controls, single points of failure, authentication and access control, administrative access, device configuration, unnecessary services, outdated firmware/software, wireless security, open ports, insecure protocols, data protection, logging and other relevant weaknesses. Explain why each significant issue matters to the selected client rather than providing a generic vulnerability list.

Task 4 – Vulnerability and Risk Assessment

Conduct a structured assessment of the existing network (Option A) or proposed/expanded network (Option B). Identify important assets, threats, vulnerabilities, existing controls and possible organizational impact. Prioritize risks using clearly defined likelihood and impact levels.

Area	Required Information
Asset/System	Affected asset or service
Vulnerability	Identified security weakness
Threat	Relevant threat
CIA Impact	Confidentiality, Integrity and/or Availability
Likelihood & Impact	Low / Medium / High
Risk Level	Prioritized risk
Existing Controls	Controls currently in place
Recommended Controls	Proposed treatment/security controls

With explicit client authorization, non-destructive activities such as vulnerability scanning, configuration review, network/service discovery, firewall-rule review or log analysis may be used. If technical testing is not permitted, use interviews, observations, architecture reviews, checklists, sanitized client information or a laboratory/simulated environment.

Ethical requirement: Unauthorized scanning, exploitation, password attacks, denial-of-service testing, destructive testing, social engineering or any activity that could negatively affect the client's systems is strictly prohibited.

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Based on the significant vulnerabilities and risks identified in Task 4, use the provided ISO 27001 Toolkit to support the risk assessment and recommend appropriate security controls and risk-treatment measures.

- Select and adapt only the ISO 27001 Toolkit materials relevant to the selected client.
- Use the toolkit to:
 - Identify and categorize important network and information assets.
 - Assess asset importance and security requirements.
 - Identify and prioritize vulnerabilities and risks.
 - Identify existing controls and security gaps.
 - Recommend appropriate risk-treatment measures.
 - Determine residual risks and implementation priorities.
- Relevant resources may include the Asset Register, Information Asset Valuation and Classification materials, Risk Register, and control-gap assessment materials.

For Option B, apply the same toolkit-based approach to the proposed network environment so that foreseeable vulnerabilities and risks are identified and treated before implementation, supporting a security-by-design approach.

For additional guidance and a better understanding of **ISO/IEC 27001:2022 and Annex A security controls**, students may refer to the official ISO resources below. Consultation of Annex A is **optional** and may be used to further support the risk assessment and security recommendations.

ISO/IEC 27001:2022 – Official ISO Page: [ISO/IEC 27001:2022 - Information security management systems](#)

Official Guidance on ISO/IEC 27001 Annex A: [ISO/IEC 27001 Auditing Practices Group – Annex A Guidance](#)

Task 6 – Secure Network Architecture (To-Be Design)

Develop a To-Be Secure Network Architecture based on the identified requirements and risks. Each significant control should address an identified vulnerability, threat or business requirement. Depending on the client, consider VLANs and segmentation, inter-VLAN controls, firewalls, DMZs, IDS/IPS, secure wireless, VPN/remote access, secure management networks, authentication/authorization, least privilege, secure protocols, endpoint protection, centralized logging/SIEM, backup and recovery, redundancy and cloud security controls. Clearly show security zones, trust boundaries and major controls in the final diagram.

Task 7 – Final Recommendations and Implementation Priorities

Provide a prioritized security improvement plan categorized as Immediate/Critical, Short-Term and Long-Term. Where relevant, consider cost, business impact, implementation complexity, scalability and operational feasibility.

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4. Required ENISS Deliverables

1. Client/network security context and selected Option A or B with justification.
2. Network requirements, critical assets and As-Is/Baseline network architecture.
3. Existing security controls and initial vulnerability identification.
4. Vulnerability and risk assessment with supporting methodology/evidence where applicable.
5. ISO/IEC 27001:2022-aligned risk register and relevant Annex A control mapping.
6. Proposed To-Be secure network architecture and justification of recommended controls.
7. Prioritized implementation recommendations.
8. Conclusion, references and appendices/supporting evidence.

5. Group Presentation and Viva

Each group must present the key findings and recommendations from the assessment, covering the client context, As-Is/Baseline network, major vulnerabilities and risks, application of relevant ISO 27001 Toolkit materials, proposed risk-treatment measures, To-Be secure network design, and prioritized security recommendations.

All members must participate and use appropriate diagrams and supporting evidence. Confidential client information must be anonymized or redacted.

The presentation will be followed by a viva to assess each student's understanding, contribution, and ability to justify the identified risks, selected security controls, ISO/IEC 27001 mappings, and proposed network design.

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Marking Scheme

Criteria	Excellent	Good	Satisfactory	Poor	Very Poor
1. Client Context, Requirements & Baseline Architecture (20 marks)	16-20: Client context and selected option are clearly justified. Critical assets, network/security requirements and existing controls are comprehensively identified. As-Is/Baseline architecture is accurate, clear and reflects the actual client environment.	13-15: Good understanding of the client and requirements. Most important assets and controls are identified. Architecture is generally accurate with minor omissions.	10-12: Adequate client analysis and basic requirements/assets identified. Architecture represents the environment but lacks some detail or clarity.	5-9: Limited client analysis. Requirements/assets are incomplete or generic. Architecture has significant omissions or inaccuracies.	0-4: Client analysis is very limited or missing. Requirements/assets are largely absent and architecture is missing or unsuitable.
2. Vulnerability & Threat Identification (10 marks)	9-10: Accurately identifies a comprehensive set of relevant, client-specific vulnerabilities and associated threats. Existing controls/security weaknesses are clearly recognized and the significance of each major vulnerability is well explained.	7-8: Identifies most important vulnerabilities and associated threats with good client relevance. Existing controls are considered, with minor omissions.	5-6: Identifies several relevant vulnerabilities and threats, but some are generic or insufficiently explained. Existing controls receive limited consideration.	3-4: Identifies only a few relevant vulnerabilities. Threat relationships and client-specific explanations are weak or incomplete.	0-2: Vulnerability/threat identification is largely missing, technically incorrect or consists mainly of unsupported generic lists.
3. Risk Assessment & Prioritization (15 marks)	13-15: Systematically assesses identified risks using appropriate CIA impact, likelihood and impact criteria. Risk levels are consistent, well justified and appropriately prioritized using suitable evidence/methodology.	10-12: Risk assessment is mostly systematic and appropriate. Likelihood, impact and prioritization are generally justified, with minor inconsistencies.	7-9: Basic risk assessment is completed. Likelihood, impact and risk levels are provided, but justification or prioritization lacks depth or consistency.	4-6: Risk assessment is incomplete or inconsistent. Risk ratings have limited justification and prioritization is weak.	0-3: Risk assessment is largely absent or technically incorrect, with little evidence supporting risk ratings or priorities.
4. ISO 27001 Toolkit Application & Risk Treatment (20 marks)	16-20: Relevant toolkit materials are appropriately selected and adapted to the client. Asset/risk/control-gap information is effectively used to develop well-justified treatments, residual risks and implementation priorities.	13-15: Relevant toolkit materials are appropriately applied and linked to major risks and treatments. Residual risks and priorities are mostly clear, with minor gaps.	10-12: Some suitable toolkit materials are used, but application is basic or partly mechanical. Treatments are generally relevant but residual risk or priorities lack depth.	5-9: Limited or poorly adapted toolkit use. Weak connection exists between toolkit outputs, identified risks and proposed treatments.	0-4: Toolkit materials are not meaningfully applied or are incorrectly used. Risk-treatment recommendations are largely unsupported or absent.
5. To-Be Secure Network Design & Implementation Recommendations (20 marks)	16-20: Presents a technically sound, secure and scalable To-Be architecture directly addressing identified requirements and risks. Security zones, segmentation, trust boundaries and controls are clearly shown and justified. Recommendations are realistic and appropriately prioritized.	13-15: Strong and mostly appropriate secure architecture. Major controls are relevant and justified. Recommendations are practical and prioritized with minor gaps.	10-12: Architecture addresses basic security requirements but lacks some depth, scalability or justification. Some controls are weakly connected to identified risks.	5-9: Design contains significant security/architectural weaknesses. Controls are incomplete or poorly justified and recommendations lack clear prioritization.	0-4: To-Be architecture is missing, unsuitable or technically unsound. Recommended controls do not adequately address identified risks.
6. Presentation, Evidence & Individual Viva (15 marks)	13-15: Presentation is professional, concise and supported by effective diagrams/evidence. All members contribute meaningfully. Viva demonstrates strong individual understanding and ability to justify vulnerabilities, risks, toolkit use, controls and design decisions.	10-12: Well-organized presentation with good supporting evidence and participation. Viva demonstrates good understanding and justification with minor weaknesses.	7-9: Main findings are communicated adequately but evidence, clarity or participation is limited. Viva demonstrates basic understanding with some difficulty justifying decisions.	4-6: Presentation is unclear or weakly supported and participation is uneven. Viva demonstrates limited understanding of assessment and design decisions.	0-3: Presentation is incomplete or poorly communicated with little evidence. Viva demonstrates very limited individual understanding or contribution.